

CoDEsk

Product Requirement Document (PRD)

Project Overview

Project Name	CoDEsk
Project Lead	Kanhaiya Tyagi
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Abstract

CoDEsk is a developer tool that provides AI-powered code review in a clean, minimal interface. A developer pastes a code snippet, selects the language, and receives structured feedback — covering issues, suggestions, and improvements — powered by an LLM. All review sessions are saved so the developer can revisit and track their past reviews.

Vision

A focused, distraction-free space where developers get instant and meaningful feedback on their code — without leaving the browser.

Goals

- Goal 1: Complete a working full-stack MVP within 3 weeks
- Goal 2: Integrate AI-powered code review via Claude/OpenAI API with structured, readable output
- Goal 3: Build a clean session history so past reviews are never lost
- Goal 4: Deploy the app publicly so it's accessible via a live link on the resume

Scope

In Scope (V1)

- Code input with syntax highlighting
- Language selector (JavaScript, Python, C, Java, HTML/CSS)
- AI review output — structured into Issues, Suggestions, Summary
- Save each review session to database
- View past sessions (history page)
- Basic error handling and loading states

Explicitly Out of Scope (V1)

- User authentication / login
- File upload or file system access
- Integrated terminal
- Team collaboration
- Custom themes
- Mobile responsiveness

Core Features

1. Code Input Panel

A clean textarea with syntax highlighting where the user pastes their code snippet. Includes a language dropdown and a "Review" button.

Field	Detail
Input type	Paste / type manually
Language support	JS, Python, C, Java, HTML/CSS
Priority	High

2. AI Review Output

After submission, the AI response is displayed in a structured, readable format — not a wall of text.

Section	Description
Summary	One paragraph overall assessment
Issues	Bugs or errors found
Suggestions	Improvements and best practices
Priority	High

3. Session History

Every review is saved to the database and accessible from a history panel or page.

Field	Detail
Stored data	Code snippet, language, AI response, timestamp
View	List of past sessions, click to expand
Priority	High

Tech Stack

Layer	Technology
Frontend	React, React Query, react-syntax-highlighter
Backend	Node.js, Express, Prisma
Database	PostgreSQL
AI	Claude API / OpenAI API
Deployment	Vercel (frontend), Render (backend)

Design Principles

- Minimal UI — two panels, input on left, output on right
- No unnecessary navigation or clutter
- Fast feedback — loading state shown immediately on submission
- Readable output — AI response rendered with clear sections, not raw text

What's Intentionally Simple

This is a V1 personal project. The goal is a fully working, deployed, well-structured codebase — not feature completeness. Every decision prioritizes finishing over expanding.